

Executive Summary

Bank Marketing Campaign Analysis

1. Project Overview

This project analyzes the Bank Marketing Dataset from the [UCI Machine Learning Repository](#), using the additional full dataset containing 41,188 customer contacts collected during direct telemarketing campaigns conducted by a Portuguese banking institution.

2. Business Objective

Identify the customer characteristics, financial profile, campaign practices, and macroeconomic conditions that are most strongly associated with successful term deposit subscriptions, and translate these findings into actionable recommendations for improving future telemarketing campaigns.

3. Business Question

Which customer characteristics, financial profile attributes, campaign strategies, and macroeconomic conditions are associated with higher subscription rates, and how can these insights be used to improve the effectiveness of future telemarketing campaigns?

4. Overall Campaign Performance

The campaign contacted 41,188 customers, of whom 4,640 subscribed to a term deposit, resulting in an overall subscription rate of 11.27%.

Metric	Value
Customers Contacted	41,188
Subscribers	4,640
Overall Subscription Rate	11.27%

The results indicate that only around one in nine contacted customers subscribed, highlighting substantial opportunities to improve campaign targeting and efficiency.

5. Key Insights

5.1. Customer characteristics strongly influence subscription behaviour

Customer demographics showed the largest differences in subscription performance.

The strongest customer segments were:

- Customers aged 60+ (39.56% subscription rate)
- Retired customers (25.23%)
- Students (31.43%)

The multidimensional analysis further showed that:

- Retired customers aged 60+ achieved 42.55% across 785 customers
- Students aged 17–29 achieved 33.43%

By contrast:

- Customers aged 40–49 recorded the lowest subscription rate (7.92%)
- Blue-collar workers recorded only 6.89%

Age and occupation therefore provide substantially stronger customer segmentation than marital status, education or financial status.

5.2. Campaign strategy had a major impact on performance

Several campaign practices were strongly associated with subscription success.

5.2.1. Contact method

Cellular communication substantially outperformed telephone calls.

Channel	Subscription Rate
Cellular	14.74%
Telephone	5.23%

5.2.2. Campaign frequency

Subscription rates were highest during the first contact attempt and generally declined after the third contact. Repeated follow-up contacts generated diminishing returns while increasing campaign costs.

5.2.3. Previous campaign outcome

Previous campaign history proved to be one of the strongest predictors of future success.

Previous Outcome	Subscription Rate
Success	65.11%
Failure	14.23%
No Previous Campaign	8.83%

Customers who had previously subscribed remained highly likely to subscribe again.

5.2.4. Campaign timing

Campaign performance also varied across the year.

The highest subscription rates occurred during:

- March
- September
- October
- December

Meanwhile, May generated the largest campaign volume but one of the lowest subscription rates (6.43%), suggesting that more activity does not necessarily produce better results.

3. Financial characteristics showed limited influence

Housing loans, and personal loans produced only small differences in subscription rates.

Examples include:

- Housing loan:
 - Yes: 11.62%
 - No: 10.88%
- Personal loan:
 - No: 11.34%
 - Yes: 10.93%

These variables appear to contribute relatively little to customer targeting compared with age, occupation and campaign history.

6. Macroeconomic conditions influenced customer behaviour

Macroeconomic conditions influenced customer behaviour, particularly during periods of lower Euribor rates and weaker labour market conditions.

Consumer Price Index and Consumer Confidence Index showed varying subscription rates, but no consistent linear relationship.

These findings suggest that economic conditions should support campaign planning rather than be used as standalone decision variables.

7. Multidimensional analysis identified the strongest customer profiles

Combining customer characteristics with campaign information produced even stronger insights.

The highest-performing segment was:

- Customers aged 60+
- Contacted via cellular
- Following a previous successful campaign

This segment achieved a 75.39% subscription rate.

Another large high-performing segment was:

- Customers aged 30–39
- Contacted via cellular
- Previous successful campaign

This group achieved a 63.37% subscription rate across 415 customers.

These results demonstrate that combining customer demographics with campaign history provides substantially better customer segmentation than analysing variables individually.

8. Business Recommendations

Based on the analysis, the following actions are recommended:

8.1. Prioritize high-value customer segments

Focus future campaigns on customers most likely to subscribe, particularly:

- Customers aged 60+
- Retired customers
- Students

This is expected to increase subscription rates while reducing unnecessary marketing effort.

8.2. Make cellular the primary communication channel

Cellular communication consistently outperformed telephone contacts across almost every campaign scenario. Future campaigns should prioritize cellular communication whenever possible.

8.3. Maximize the effectiveness of the first few contacts

Subscription rates were highest during the earliest customer interactions.

Rather than repeatedly contacting the same customers, marketing teams should focus on improving the quality of initial conversations.

8.4. Use previous campaign history for customer targeting

Previous campaign success proved to be one of the strongest predictors of future subscriptions.

Campaign history should therefore become an important feature in future customer selection strategies.

8.5. Optimize campaign timing

Campaign resources should be concentrated during historically stronger-performing months rather than simply increasing campaign volume.

8.6. Focus on the variables that matter most

Marketing segmentation should primarily rely on:

- Age
- Occupation
- Previous campaign outcome
- Contact method

Variables such as housing loans, personal loans and marital status can provide supporting information but should not drive targeting decisions.

8.7. Consider macroeconomic conditions during campaign planning

Although macroeconomic indicators are not direct predictors of customer behaviour, incorporating them into campaign planning may improve timing and resource allocation.

9. Conclusion

This analysis demonstrates that improving telemarketing performance is not about contacting more customers it is about contacting the right customers, through the most effective communication channel, at the optimal time.

The analysis identified several factors that are consistently associated with higher subscription rates, including:

- Older customers, particularly those aged 60+
- Occupation, especially retired customers and students
- Cellular as the preferred communication channel
- Early campaign contacts
- Previous campaign success
- Campaign timing, particularly March, September, October, and December
- Broader macroeconomic conditions, especially lower Euribor rates and weaker labour market conditions

These findings provide a practical framework for improving customer targeting, optimizing campaign strategy, and supporting more informed marketing decisions. By shifting from a high-volume telemarketing approach to a targeted, data-driven strategy, the bank can improve subscription rates, reduce unnecessary marketing costs, and allocate campaign resources more efficiently.